This is single document summarization application using MapReduce Architecture.

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
# !pip install langchain transformers sentence-transformers
!pip install langchain transformers accelerate bitsandbytes pypdf sentence-transformers faiss-cpu PyPDF2 tiktoken
!pip install langchain-community
!pip install langchain-huggingface
!pip install -U transformers
!pip install -U transformers langchain sentence-transformers
```

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Successfully installed langchain-huggingface-1.2.1
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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.
```

```
## Document Loader and Chunking
from langchain_community.document_loaders import PyPDFLoader
from langchain_text_splitters  import RecursiveCharacterTextSplitter
from langchain_core.documents  import Document
```

```python
from sentence_transformers import SentenceTransformer
import PyPDF2


# ========================
# 1. Read PDF
# ========================
def read_pdf(file_path):
    pdf_reader = PyPDF2.PdfReader(file_path)
    text = ""
    for page in pdf_reader.pages:
        text += page.extract_text() + "\n"
    return text

pdf_text = read_pdf("/content/drive/MyDrive/LangChain Learning/document/responsible_finance.pdf")

# ========================
# 2. Semantic Chunking
# ========================
text_splitter = RecursiveCharacterTextSplitter(
    chunk_size=500,   #conventional size is used
    chunk_overlap=50,
    length_function=len
)
chunks = text_splitter.split_text(pdf_text)

# ========================
# 3. Store chunks as Documents with metadata
# ========================
documents = []
for i, chunk in enumerate(chunks):
    doc = Document(
        page_content=chunk,
        metadata={
            "chunk_id": i,
            "source": "responsible_finance.pdf"
        }
    )
    documents.append(doc)

print(f"Total chunks created: {len(documents)}")
print(documents[0].metadata, documents[0].page_content[:100], "...")

document_sections = []
for document in documents:
    document_sections.append(Document(page_content=document.page_content))
print('Number of Chunks: ', len(document_sections))
document_sections[0], type(document_sections[0])
```

```
Total chunks created: 152
{'chunk_id': 0, 'source': 'responsible_finance.pdf'} Early microfinance pioneers created the sector
with the aspiration to improve poor people’s
lives. ...
Number of Chunks:  152
(Document(metadata={}, page_content='Early microfinance pioneers created the sector \nwith the aspiration to improve poor
people’s \nlives. There were huge swathes of “white space” where low-income households had little or no \naccess to formal
finance. This challenge, in turn, \ndemanded sustainable and scalable models for \ndelivering adequate financial services to
poor \npeople. \nThis is the essence of the double bottom line in \nmicrofinance: a social commitment to benefiting
\nclients married with a financial commitment to'),
 langchain_core.documents.base.Document)
```

```python
# Imports
from langchain_core.prompts import PromptTemplate
from langchain_classic.chains import LLMChain, MapReduceChain, load_summarize_chain
from langchain_community.llms import HuggingFacePipeline
from transformers import pipeline
from sentence_transformers import SentenceTransformer
from typing import List

# ========================
# 1. Setup LLM (HuggingFace free model)
# ========================
hf_pipeline = pipeline(
    task="text-generation",
    model="google/flan-t5-base",  # Free model, smaller & fast
    max_new_tokens=256,
```

```
    temperature = 0.1,
    max_length=None
)

```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json:       1.40k/?  [00:00<00:00,  120kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faste
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
model.safetensors:  100%                                                990M/990M  [00:10<00:00,  207MB/s]
Loading weights:  100%                                                  282/282  [00:00<00:00,  5961.09it/s]
The tied weights mapping and config for this model specifies to tie shared.weight to lm_head.weight, but both are present in
generation_config.json:  100%                                             147/147  [00:00<00:00,  3.95kB/s]
tokenizer_config.json:       2.54k/?  [00:00<00:00,  67.3kB/s]
spiece.model:  100%                                                     792k/792k  [00:00<00:00,  4.06MB/s]
tokenizer.json:       2.42M/?  [00:00<00:00,  24.3MB/s]
special_tokens_map.json:       2.20k/?  [00:00<00:00,  41.0kB/s]
Passing `generation_config` together with generation-related arguments=({'max_length', 'max_new_tokens', 'temperature'}) is d
The model 'T5ForConditionalGeneration' is not supported for text-generation. Supported models are ['PeftModelForCausalLM', 'A
/tmp/ipykernel_904/507144027.py:20: LangChainDeprecationWarning: The class `HuggingFacePipeline` was deprecated in LangChain
  llm = HuggingFacePipeline(pipeline=hf_pipeline)
```

```
# ========================
# 2. Define Map and Reduce Prompts
# ========================
map_prompt = PromptTemplate(
    input_variables=["text"],
    template="""  Summarize the following text in 3 concise sentences.\n\n {text}"""
)

reduce_prompt = PromptTemplate(
    input_variables=["text"],
    template="""Combine the following summaries into a clear and concise final summary:\n\n{text}"""
)

# ========================
# 3. Define Map and Reduce Chains
# ========================
map_chain = LLMChain(llm=llm, prompt=map_prompt)
reduce_chain = LLMChain(llm=llm, prompt=reduce_prompt)

# ========================
# 4. MapReduceChain
# ========================
summary_chain = load_summarize_chain(
    llm,
    chain_type="map_reduce", ## As MapReduce Architecture Used
    map_prompt=map_prompt,
    combine_prompt=reduce_prompt
)

# ========================
# 5. Run MapReduce Summarization
# ========================
summary_result = summary_chain.invoke(document_sections[5:10]) ## Passing five paragrams to save resources
print("=== FINAL SUMMARY ===")
print(summary_result)
```

```
/tmp/ipykernel_904/3711912946.py:17: LangChainDeprecationWarning: The class `LLMChain` was deprecated in LangChain 0.1.17 and
  map_chain = LLMChain(llm=llm, prompt=map_prompt)
/usr/local/lib/python3.12/dist-packages/langchain_core/language_models/base.py:354: UserWarning: Using fallback GPT-2 tokeniz
  return len(self.get_token_ids(text))
```

tokenizer_config.json: 100% 26.0/26.0 [00:00<00:00, 1.27kB/s]

vocab.json: 1.04M/? [00:00<00:00, 4.81MB/s]

merges.txt: 456k/? [00:00<00:00, 1.21MB/s]

tokenizer.json: 1.36M/? [00:00<00:00, 1.98MB/s]

```
=== FINAL SUMMARY ===
{'input_documents': [Document(metadata={}, page_content='delivery. We need to better manage the trade-\noffs to ensure that p
```

```
print(summary_result['output_text'])
```

```
Combine the following summaries into a clear and concise final summary:

 Summarize the following text in 3 concise sentences.

 delivery. We need to better manage the trade-
offs to ensure that products and practices are
sound and client welfare and institutional viability
are not put at risk. Scale and sustainability goals
remain legitimate and, indeed, critical goals, but they must be supported with the necessary
infrastructure, such as credit bureaus, and
public goods, such as transparency regulation.
2
In sum, we need to rebalance the double bottom line as our collective goal shifts from simply filling

 Summarize the following text in 3 concise sentences.

 in the white space to developing a healthy and
responsible market.
Against this backdrop of high aspiration and new
challenges, the microfinance field has embraced
the need for a proactive responsible finance agenda
and accelerated its efforts to implement meaningful
improvements in products and practices. There is a growing consensus that providers of financial
services to the poor must adhere to a standard
of care that minimizes risks for their customers,

 Summarize the following text in 3 concise sentences.

 who typically have low and variable incomes, little
margin for error in financial decision making, and
limited formal education and exposure to formal
finance.
3
In this paper, we define what we mean by responsible finance, both as an end-state vision and
in terms of a pragmatic focus on client protection
and social performance management to help
achieve our goal. This paper explores the state
of responsible finance knowledge and practice,

 Summarize the following text in 3 concise sentences.

 of responsible finance knowledge and practice,
with a focus on three mutually reinforcing client
protection strategies: Responsible Finance:
Putting Principles to Work
1 See Chen, Rasmussen, and Reille (2010) and CGAP (2010).

 Summarize the following text in 3 concise sentences.

 2 Transparency is a general problem for all financial services, since their features and quality are difficult for consumers
```

Double-click (or enter) to edit